

### Introduction to ADaM data structures

- In clinical trials, data specified in the protocol is collected using case report forms
- The data collected using case report forms is called raw data
- Raw data is then transformed into CDISC compliant SDTM datasets
- SDTM data is used as input to create analysis datasets
- Analysis datasets compliant with CDISC ADaM standard are called ADaM datasets
- CDISC ADaM standard has three standard dataset structures
  - ADSL- Subject Level Analysis Dataset
  - BDS - Basic Data Structure
  - OCCDS - Occurrence Data Structure

### Introduction to BDS structure

- The typical record structure in a BDS dataset is 'one or more records per subject, per analysis parameter, per analysis timepoint'
- An easy phrase to remember the structure of BDS is 'a dataset that is used to store the results of tests'
- A 'Result' is formally called an 'analysis value'
- A 'test' is formally called as 'analysis parameter'
- A 'test' could be conducted at different timepoints. Hence, we have the concept of 'analysis timepoint' as well.
- So, there are core variables prescribed in ADaM standard for 'analysis parameter' and 'analysis value'
- Variables for 'analysis parameter' are PARAMCD, PARAM, PARAMN
- Important variables for 'analysis result' are AVAL, AVALC.
- Variables for baseline value (BASE), change from baseline (CHG), and percent change from baseline (PCHG) are also considered as analysis values
- Similar to variables for 'analysis parameter' and 'analysis value', there are standard variables prescribed for 'analysis timepoint'
- Variables for 'analysis timepoint' include AVISIT, AVISITN, ADT, ATM, ADTM, ADY
- Here is a visualization of some variables for data related to vital signs

|   | USUBJID  | PARAMN | PARAMCD | PARAM                                   | AVISIT | ADT       | AVAL |
|---|----------|--------|---------|-----------------------------------------|--------|-----------|------|
| • | CSG-1001 | 1      | STSBP   | Sitting Systolic Blood Pressure (mmHg)  | Week 1 | 15APR2003 | 122  |
|   | CSG-1001 | 2      | STDBP   | Sitting Diastolic Blood Pressure (mmHg) | Week 1 | 15APR2003 | 74   |

### Introduction to rules governing addition of new parameters

- When creating analysis datasets, we typically run into the question of how to structure the derived information - whether to add rows or columns.
- ADaM implementation guide provides certain rules that guide on when to add the information as rows and when to add the information as columns
- Within the concept of rows - we end up with two options
  - Add the derived information as rows within existing parameter
  - Add the derived information as rows as a new parameter
- Of the 6 rules available, 3 rules are related to addition of new parameters

2. **Add additional derived data as needed for the analysis.** The second step consists of adding derived rows and columns based on the initial set of ADaM dataset records and columns. *The rules below govern this step.* These rules are further described and illustrated in the remaining subsections of this section.

**Rule 1:** A parameter-invariant function of AVAL and BASE on the same row that does not involve a transform of BASE should be added as a new column.

**Rule 2:** A transformation of AVAL that does not meet the conditions of Rule 1 should be added as a new parameter, and AVAL should contain the transformed value.

**Rule 3:** A function of one or more rows within the same parameter for the purpose of creating an analysis timepoint should be added as a new row for the same parameter.

**Rule 4:** A function of multiple rows within a parameter should be added as a new parameter.

**Rule 5:** A function of more than one parameter should be added as a new parameter.

**Rule 6:** When there is more than one definition of baseline, each additional definition of baseline requires the creation of its own set of rows.